



USER PERSONA · SCRATCH MATCH

Andre Yap

Developer

Student-engineer maintaining Scratch Match end-to-end.

DEMOGRAPHICS

AGE 21

AFFILIATION Group Sigma · CIS 2205, USC

STACK Next.js · SQLite · Qdrant · Ollama

ENVIRONMENT Local, offline-capable, no cloud keys

TECH COMFORT Builds, deploys, and debugs the system

BACKGROUND

Andre is one of the student engineers behind Scratch Match. He cares about the matcher actually being usable on commodity hardware the city can buy outright. Every dependency is open-source. Every model runs locally. The architecture is intentionally boring so a future maintainer (probably another student) can pick it up in a weekend.



CORE NEED

As a Developer, I want to Maintain and tune the matcher engine, keep the system offline and auditable, and ship updates without any cloud dependencies.

Goals

- › Keep the deployment story to a single docker compose up.
- › Make the matching logic legible to a junior who joins later.
- › Ship improvements without needing an internet connection.

Frustrations

- › Most modern AI tooling assumes a paid cloud account.
- › Vendor lock-in disguised as developer convenience.
- › Tutorials that hide the moving parts behind one-line abstractions.



System Sketch

A view from the Scratch Match prototype that this persona touches.